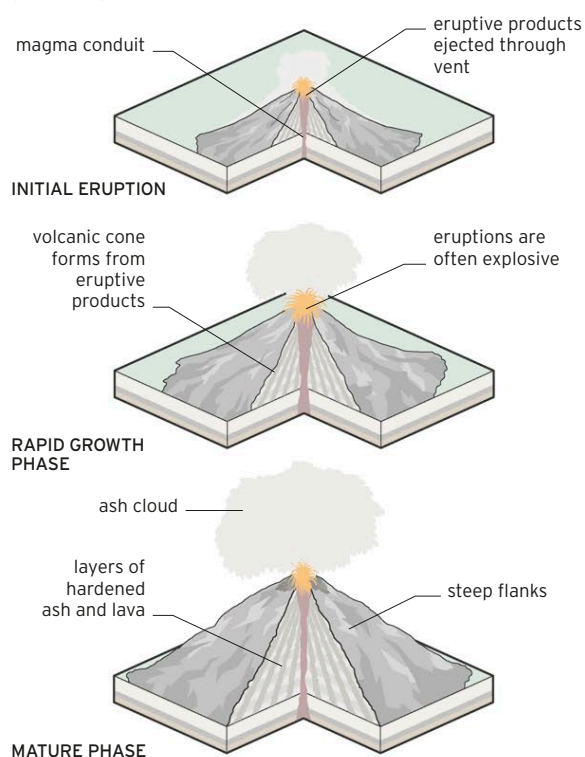


Inside a STRATOVOLCANO

There are two main types of large volcano—shield volcanos and stratovolcanos. A shield volcano slopes gently, whereas a stratovolcano has steep sides and a conical shape built up from layers of eruptive products. The reason for these differences lies in the types of lava they emit. A shield volcano produces relatively runny lava that flows a long way before solidifying. A stratovolcano, such as Etna, also produces more viscous, water-rich lava that does not flow far. Instead, the lava tends to solidify around, and can sometimes block up, the volcano's main vent, or vents. As a result, much of the material erupted by a stratovolcano consists of fragmented solids—ash, pumice, and cinders—that are blasted into the air and later fall to the ground as the volcano clears its main vent. This helps explain not just the structure and shape of stratovolcanoes but also their long-term behavior, with phases of violent activity interspersed with quiet periods, lasting from a few years to several thousand years.

THE EVOLUTION OF A STRATOVOLCANO

Stratovolcanoes grow from the accumulation of their own eruptive products: solidified lava, ash, and cinders. The volcano initially grows rapidly, as eruptions add a lot of material relative to its existing size. A mature volcano gains height more slowly, partly due to erosion.



▷ FLANK CRATERS

Etna's flanks are covered in hundreds of secondary cones, vents, and craters, many of them now inactive. This is one of a pair called the Silvestri craters, dating from an eruption in 1892. It is 360 ft (110 m) across.



Ash cloud
An ash cloud is formed by gas and magma that is explosively fragmented into tiny particles as it is blasted into the air.

Solidified lava
Etna's surface is covered in solidified lava of various ages.

directly underneath
Etna are layers
of ancient volcanoes
that form the
foundation for the
present-day cone

▷ LAVA FLOWS

Some eruptions occur from fissures that open up quite low down on the volcano's flanks. These can produce huge flows of lava that destroy agricultural land.



Lithospheric mantle
Made of the coarse-grained igneous rock peridotite, the lithospheric mantle underlies the crust.